DA2 Hotspot Detection: Feedback (G1–G19)

Common to all groups

- Report mean ± std over at least 3–5 seeds, and use a statistical test when claiming that one method beats another.
- Compare against published ICCAD-12 results (hotspot recall, false alarms, F1), not only your own baseline.
- Select models, thresholds and hyperparameters on a validation split only. Use the test set once, at the end.
- Include a work allocation table, a real limitations section and a release-ready description of code and seeds.
- Every claim needs a cited or measured basis, and each citation must support the sentence it follows.
- Keep a minimum of 30 – 35 references and include them by citing in the introduction/literature review portion of the paper.

G1 – Multi-representation learning

- Reconcile the baseline numbers that differ between Tables II and III, and fix the parameter count, which is inconsistent with the stated architecture.
- Choose the best representation on validation data, not the test set.
- Run a true ablation: leave-one-representation-out, pairwise fusion, and late or attention fusion instead of only channel concatenation.
- Explain, with evidence, why fusion underperforms (scale or information mismatch).
- Remove the submission-status page and add the work table.

G2 – DCT-enhanced detection

- Use the same validation protocol for all four settings.
- Ablate the number of DCT coefficients, the zig-zag scheme and the block size.
- Add significance testing across seeds.
- Compare with published DCT or frequency-domain hotspot work.
- Add limitations, future work and the work table.

G3 – GA feature selection

- Report on the full test set, or report both the full and subsampled sets.
- Benchmark the GA against other selectors (mutual information, L1/RFE, PCA) and another metaheuristic.
- Repeat GA runs and show selected-feature stability across seeds.
- Report GA search cost against the inference saving.
- Interpret what the selected features represent.

G4 – Encoding-aware deep learning

- Give confusion matrices for all benchmarks, not just B3.

- Diagnose the raster collapse on B3/B4 (threshold, epochs, class weights) so the baseline is fair.
- Add more encodings (density variants, DCT, learned embeddings).
- Back the "why encoding matters" explanation with analysis such as feature visualisation or information-loss measures.
- Add multi-seed results and the work table.

G5 – CNN features with classical ML

- Use the common baseline (same parameters and loss), and separate the effects of backbone, classical head and focal loss.
- Include Decision Tree and Logistic Regression.
- Report calibration and threshold sensitivity.
- Report compute cost, since two-stage pipelines add latency.
- Add confidence intervals.

G6 – Heterogeneous ensemble

- Use the common baseline, or justify the 323k-parameter CNN.
- Quantify diversity (disagreement, Q-statistic) to explain why the ensemble helps.
- Make stacking leakage-safe with out-of-fold predictions.
- Report ensemble latency and memory.
- Do the error-pattern analysis on actual clips.

G7 – Two-stage detector

- Make Stage 1 genuinely fast; the 3.3M-parameter network is not.
- Train Stage 2 on hard or ambiguous samples, not generically.
- Sweep the routing band, calibrate confidences and report the cost–accuracy curve.
- Measure timing on all benchmarks and compare against single-stage at equal compute.
- Use IEEE format and add the work table.

G8 – GA-optimized CNN

- Remove test-set fitness and use validation fitness.
- Use a much larger GA budget and repeat runs.
- Add a parameter-importance analysis (per-gene sensitivity or ANOVA).
- Compare with Bayesian optimization or random search at equal budget.
- Evaluate on all five benchmarks, add specificity and clean up the confusion-matrix table.

G9 – Cost-sensitive detection

- Add the work table.
- Tie the cost ratio to a real verification-cost model (cost of a false alarm versus a missed hotspot).
- Compare against other imbalance methods.
- Frame the negative or mixed findings as the paper's contribution.
- Discuss reproducibility and any code release.

G10 – Focal / hard-example learning

- Complete the γ and hard-fraction ablations.

- Run OHEM on all benchmarks with the same split.
- Fix the "improves on all five" claim, since B2 balanced accuracy fell.
- Define the difficult-sample subset and report per-difficulty recall.
- Compare focal loss against class weighting and resampling.

G11 – Data balancing and augmentation

- Add a validation split and equalise gradient updates across strategies.
- Add seeds.
- Test combined strategies and PR-AUC.
- Justify layout-valid augmentations physically.
- Explain mechanistically why undersampling fails.

G12 – Uncertainty-aware detection (needs the most work)

- Run per benchmark on the official test sets, not a merged corpus, and fix the confusion-matrix size.
- Produce a real three-way output (hotspot, non-hotspot, referral) and a coverage–risk curve.
- Report the false-negative rate among accepted samples and the referral rate versus threshold.
- Set the threshold on validation data, not the test set.
- Compare MC-dropout against softmax confidence, entropy and deep ensembles.
- Add calibration (ECE, reliability diagram).
- Give the baseline a confusion matrix.
- Fix citation misuse and remove the "exponentially faster" claim.
- Quantify Grad-CAM beyond qualitative figures.

G13 – Multi-objective GA for a lightweight CNN

- Specify the baseline architecture, search space, population, generations and operators.
- Report larger Pareto fronts with hypervolume.
- State the rule for choosing the final model.
- Measure latency instead of using proxies.
- Compare against pruning or hand-designed lightweight CNNs, and repeat runs.

G14 – Accuracy–latency optimization

- Broaden the model families.
- Make latency controlled (warm-up, batch size, hardware, repeated timing) and add memory or throughput.
- State an explicit operating-point rule.
- Run the resolution ablation on all benchmarks.
- Add significance testing and a clearer novelty framing.

G15 – Cross-benchmark generalization

- Add a data-driven explanation of transfer (class balance, pattern statistics, MMD or embedding distance).
- Separate architecture effects from the generalisation question.
- Add an ablation and seeds, and fix the 64×64 versus 128×128 inconsistency.

- Remove template remnants and ** markup, fix typos and citation numbering.

G16 – Limited-data adaptation

- Use more source benchmarks and the full source-to-target grid.
- Increase seeds, since 5% target data holds very few hotspots.
- Compare with other adaptation methods (domain adaptation, augmentation).
- Report specificity and state the validation protocol.
- Temper the novelty claim for ICCAD-3 to ICCAD-5, which is a known transfer pair.

G17 – Robustness to layout perturbations

- Use a separate validation set, not the test set.
- Run the sweep on the full test set and for both clean-trained and robust models.
- Make perturbations lithography-aware (edge placement, CD bias).
- Explain the non-monotonic trends.
- Fix Table I and add a robust-training ablation.

G18 – Explainable hotspot detection

- Finish the work and drop the progress-report framing.
- Use a fair baseline and keep benchmarks separate.
- Add faithfulness metrics (deletion and insertion curves, sanity checks).
- Check explanation stability across seeds.
- Validate highlighted regions against design-rule or expert judgement, and add the work table.

G19 – ML-based pre-screening and verification referral

- Select thresholds on validation data.
- Use calibrated or genuine uncertainty estimates, not raw sigmoid scores.
- Compare with the baseline at a matched false-negative rate.
- Make referral figures consistent (the ~92% and 96.56% quotes).
- Report the referral queue's cost saving as workload only, unless simulation evidence exists.